

Automated Budget Tool – MVP Specification

1. Problem Statement

Traditional budgeting tools require frequent manual input and review, which leads to low engagement and abandonment. Users often fail to track spending in real time, reducing their ability to make informed decisions during the month.

This project aims to build an automated budgeting tool that continuously tracks spending, categorises transactions with minimal user effort, and provides clear, timely feedback so users can adjust behaviour before overspending occurs.

2. Ground Rules

1. This is a first version (MVP) of the tool.
 2. The author is the sole user of the tool.
 3. The tool requires an Open Banking connection.
 4. The user must have a bank account compatible with the chosen Open Banking provider.
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3. User Requirements

Access & Authentication

1. The user will access the tool via a web application using a URL.
2. The tool must be usable on electronic devices (desktop, tablet, mobile).
3. For V1, authentication will rely on Open Banking authorisation and basic access protection (no full user account system).

Setup & Budgeting Period

4. During initial setup, the user can define a monthly budgeting period using one of the following:
 - A fixed day each month, or
 - An income-to-income period defined by a user-selected income date.
5. The budgeting period runs from the selected start date to the next occurrence of that date.
6. The user can change the budgeting period later; the system must adapt without data loss.

Account & Transactions

7. The user can view the following account details:

- Current account balance (at time of opening the tool).
 - Starting balance at the beginning of the active budgeting period.
8. The user can view all transactions within the active budgeting period.
 9. Historical budgeting periods must be viewable in read-only mode.

Categories & Limits

10. During setup, the user can create spending categories to track.
 11. The user can edit, add, or remove categories after setup.
 12. The user must set a spending limit for each category.
 13. Every transaction must belong to a category; a default category called “Other” will be used initially.
 14. The user can manually reassign transactions to different categories at any time.
 15. The user can view current spending per category against its limit.
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4. System Requirements

Platform

1. The tool will be implemented as a web application.

Security (V1 Scope)

2. The tool will use HTTPS for all communication.
3. Authentication and secure access to banking data will be handled by the Open Banking provider.
4. No banking credentials will be stored by the tool.

Banking Data

5. The tool will access transaction data via an Open Banking provider.
6. Transaction data will be fetched when the tool is opened.
7. If fetching fails, the user can manually trigger a refresh.

Caching & Persistence

8. The tool will cache transaction data for the active budgeting period to reduce repeated API calls.
9. Cached data will be refreshed only when explicitly requested or when the budgeting period changes.
10. The tool will persist transaction data, category assignments, limits, and period metadata across sessions.
11. Persistent storage will act as the source of truth; Open Banking is used only as a data source.
12. Duplicate transactions must be handled safely when data is re-fetched.

Storage

13. For V1, persistent storage will be implemented using Google Drive via the Google Drive API.
 14. Data will be stored as structured files (e.g. JSON) and must be readable and exportable for future use.
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5. Explicit Non-Goals for V1

- Multi-user support
 - Multiple bank accounts
 - Joint accounts
 - Predictive analytics or spending forecasts
 - Machine learning–based categorisation
 - Push notifications or real-time alerts
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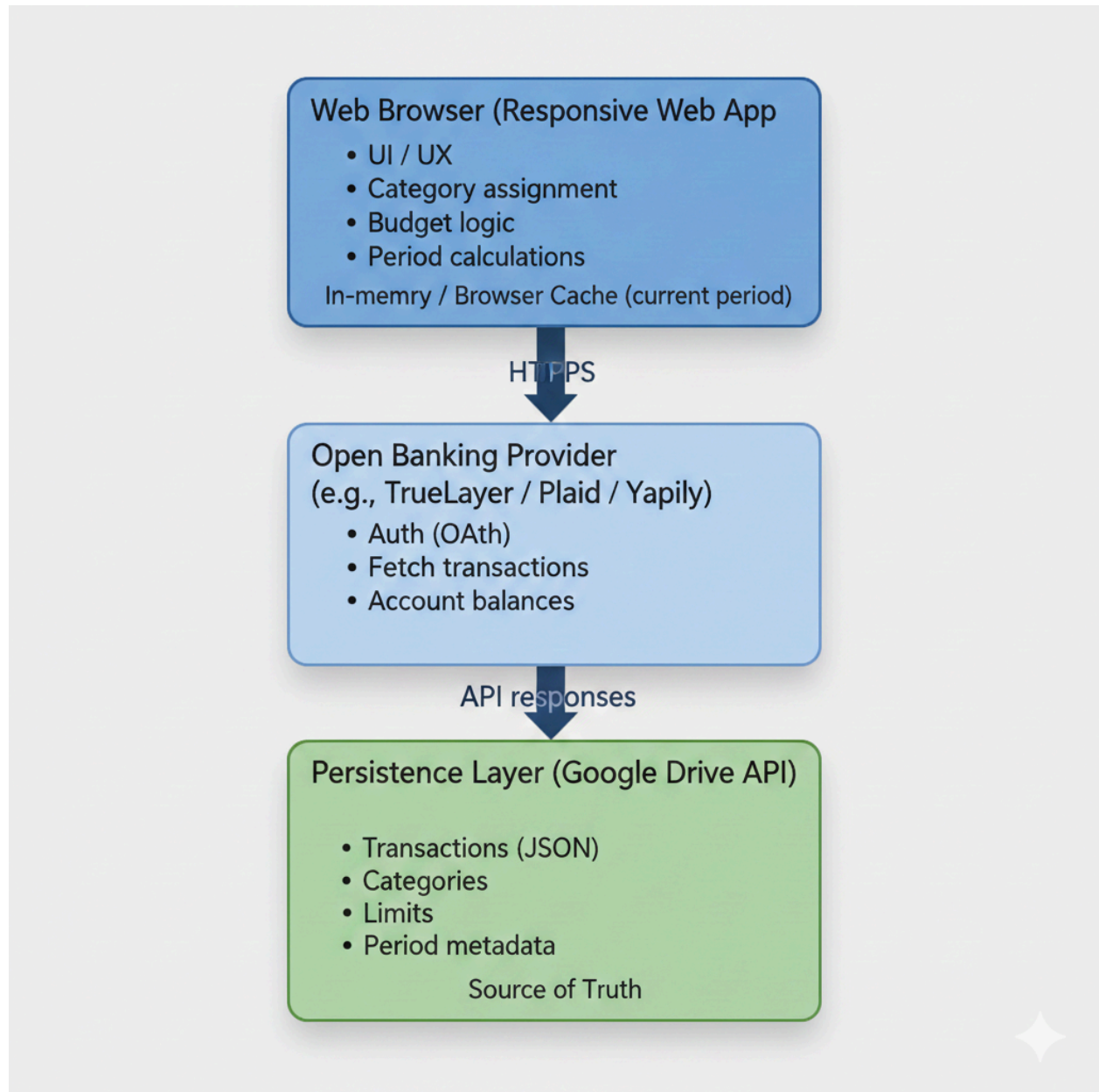
6. Future Enhancements (Out of Scope for V1)

1. Learning repeat transactions based on historical user categorisation.
 2. Automatic category assignment using previously labelled data.
 3. Notifications when category spending reaches 50%, 75%, and 100% of the set limit.
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7. Architectural Assumptions

- The tool maintains a local copy of transaction data independent of the Open Banking provider.
- Historical data is retained to support user trust, corrections, and future intelligent features.
- All design decisions prioritise simplicity, correctness, and ease of iteration for a single-user MVP.

8. Architecture



V1 Data Model

User

- user_id (string / UUID)
- created_at (timestamp)

Account

- account_id (string)
- user_id (foreign key)
- bank_name (string)
- account_name (string)
- currency (string)
- created_at (timestamp)

BudgetPeriod

- period_id (string)
- user_id (foreign key)
- account_id (foreign key)
- start_date (date)
- end_date (date)
- starting_balance (number)
- period_type (enum: FIXED_DATE | INCOME_ANCHORED)
- anchor_date (date)
- created_at (timestamp)

Category

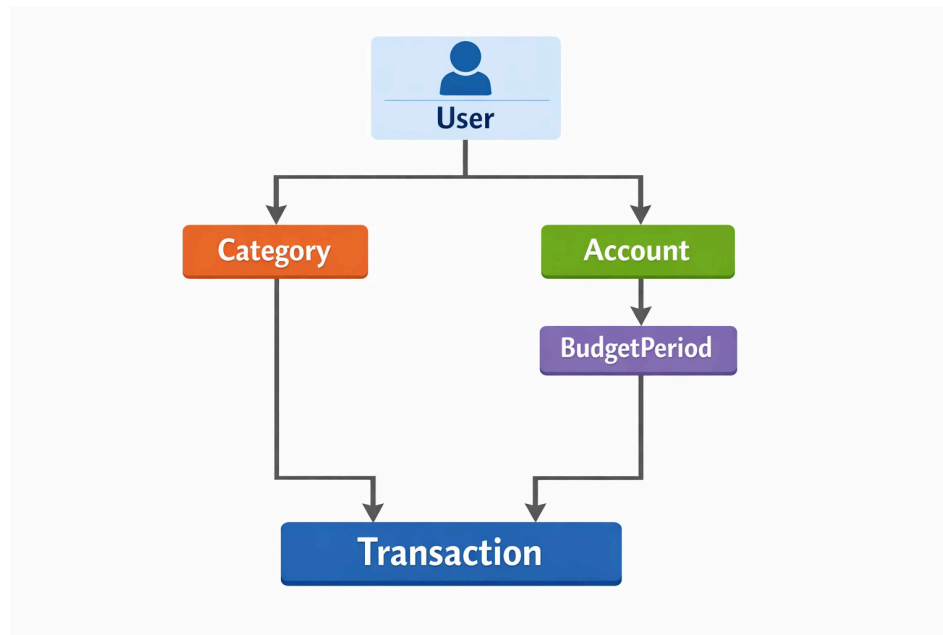
- category_id (string)
- user_id (foreign key)
- name (string)
- monthly_limit (number)
- is_default (boolean) // true for "Other"
- created_at (timestamp)
- archived_at (timestamp | null)

Transaction

- transaction_id (string) // internal ID
- external_id (string) // Open Banking ID
- account_id (foreign key)
- user_id (foreign key)
- period_id (foreign key)
- date (date)
- amount (number)
- merchant_name (string)
- description (string)

- category_id (foreign key)
- original_category (string | null)
- is_manual_override (boolean)
- created_at (timestamp)
- updated_at (timestamp)

Data Relationship



V1 Status: COMPLETE (Core MVP delivered)

V1 delivers an end-to-end automated budgeting loop using Google sign-in + Google Drive as storage, with real bank transactions imported via TrueLayer. The app supports: user/account bootstrap (“setup”), category management with a default catch-all category, an active budget period model (fixed day-of-month), transaction refresh with de-duplication, and a dashboard that shows period dates, starting balance (MVP approximation), and per-category spend/remaining. Transactions can be reviewed and manually recategorised, preserving manual overrides. TrueLayer tokens are persisted to Drive and refreshed automatically; users can reconnect via a Settings “Connect/Reconnect Bank” flow when needed. Spending calculations exclude income from spend totals (income remains visible in the transactions list for future net/cashflow features). Merchant display was removed for MVP and descriptions are cleaned for readability. V1 is deployed on Vercel and operates with production (live) TrueLayer credentials and OAuth redirect configuration.